# LATER GATOR

## SPRINT 2 PRESENTATION

### PREPARED FOR
CEN3031 – Introduction to Software Engineering
FALL 2025 | University of Florida
Professor Lisha Zhou

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Cover image created with ChatGPT (OpenAI).

## Presentation Overview

The following pages provide the slides and corresponding speaker notes for Team Later Gator's *Final Presentation*. These materials cover ***Later Gator*** from the perspective of a real user, demonstrating everything our prototype can currently do, from signing in, customizing settings, and managing app limits to experiencing the full Pomodoro study mode. We also highlight the technical decisions behind these features, the areas of technical debt we intentionally postponed, and the stretch goals we identified for future development. We conclude with the issues we faced, what we learned from them, and the next steps we would take if we continue building ***Later Gator*** beyond this course.

A recording of the full presentation can be viewed at the following unlisted YouTube link: https://youtu.be/nLUPudLmb14.

Slide 1


**SAM:** "We are **Team Later Gator**, and today we're presenting our progress on the project. --

I'm Sam – *Project Manager*, responsible for helping define the product vision, prioritize product backlog features, and ensure the team develops feature based on the user's needs. "

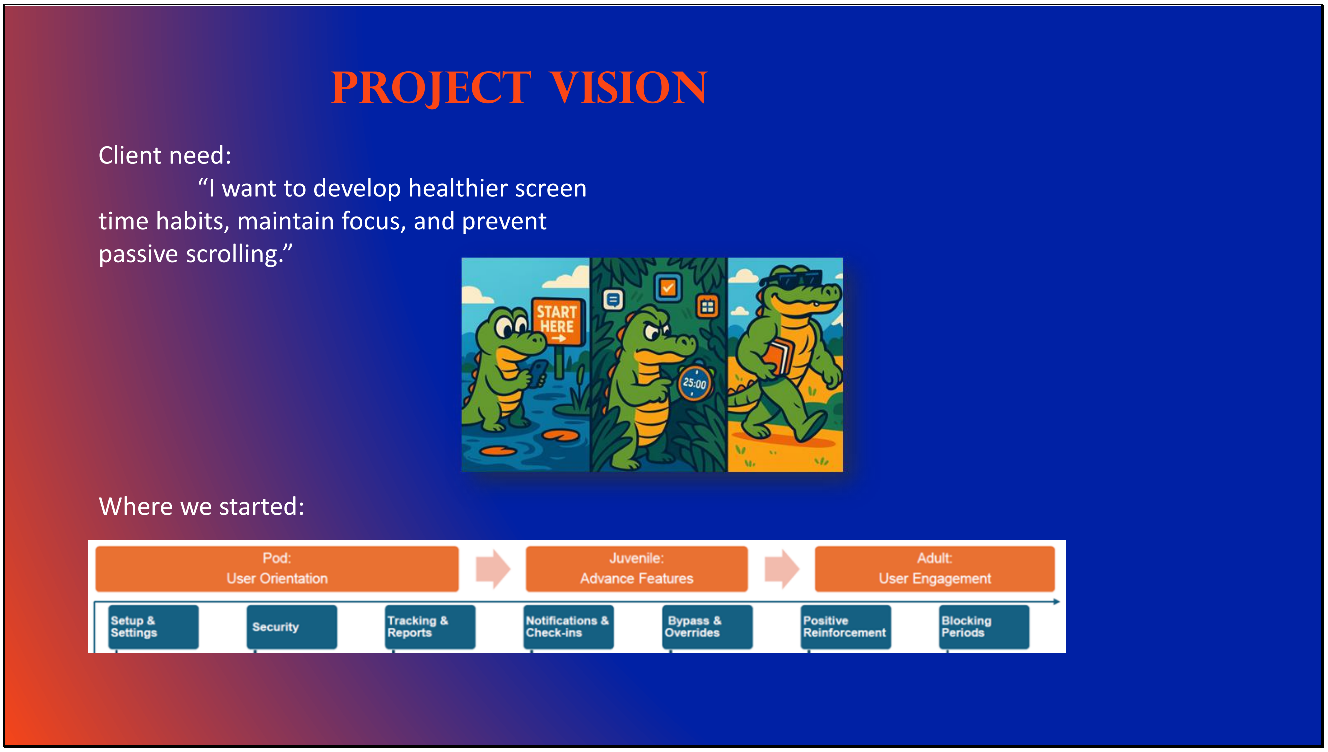
**OTHERS:**

I'm **Madison** – *Scrum Master*, responsible for maintaining our Jira and ensuring our sprints go according to plan

I'm **Danielle** – *Developer*, responsible for … Ensuring correct coding environment and tools and that product development best practices are followed.

I'm **Carrie** – *Developer*, responsible for … I work with Danielle on keeping the team well resourced for our product development.

Slide 2




**SAM:**

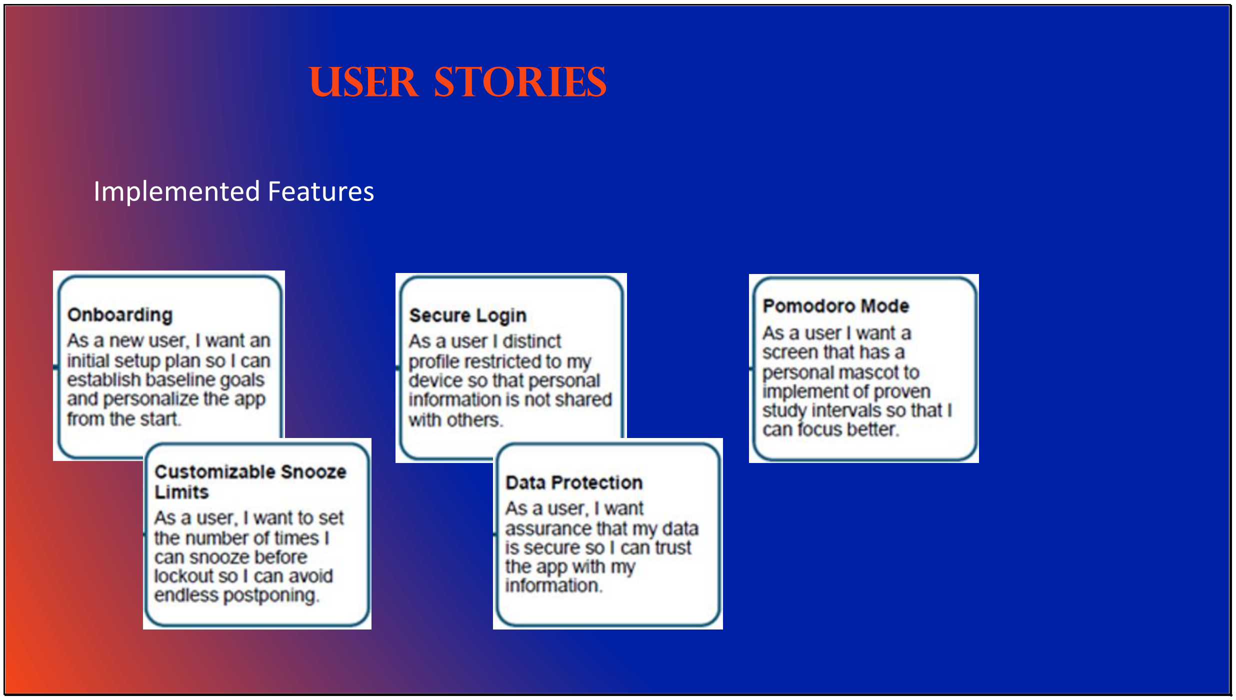
“Our client need is: ‘I want to develop healthier screen time habits, maintain focus, and prevent passive scrolling.’

To reach these goals, we developed a set of user stories from a user's point of view. These stories showed that users need support across 4 major areas: awareness, control, motivation, and mindfulness as well as an app the that is reliable and secure. Awareness comes from tracking and reports that help users understand where their time goes. Control comes from blocking periods, snooze limits, and emergency overrides that reduce distractions without feeling restrictive. Motivation comes from rewards, streaks, and positive reinforcement to help users stay committed. And mindfulness comes from notifications, check-ins, and reflections that keep users conscious of their goals throughout the day.”

“Our prototype aims to lay the foundation for these long-term goals, There are more user stories to complete In future development. we plan to connect more tracking to our database and provide tracking report, as well as work on a possible gamification system to positively enforce user goals.”

Now I'll walk through the features we’ve actually implemented so far .
**[NEXT SLIDE]**

Slide 3




**SAM:**

We needed an excellent foundation to build from. Briefly list features we did implement…

So in this early version, we focused on implementing the structural pieces that the full system will rely on. First, setup our database structure. We set up secure login using Firebase to ensure data protection and give users a reliable authentication process. We created a functional home screen that navigates to the settings page and to our Pomodoro style study mode.

In the settings screen, we implemented the ability to update the user name, adjust permissions, choose apps for blocking, and configure snooze limits and app time limits. These settings represent the early framework for the control and customization features described in the user stories. In terms of onboarding, we do not have a fully featured onboarding yet, but we will show the elements that have been implemented.

We also built the full front-end for our Pomodoro study mode which is proto-type read. This includes the complete UI, the study and break screens, and the ability for users to input the number of sessions they want to complete. However we pushed the backend integration with the database into future sprints

Pass to Danielle for walkthrough
**NEXT SLIDE**

Slide 4


**DANIELLE:**

Now, we will walk through Later Gator exactly as a user would experience it.

Google Sign-In

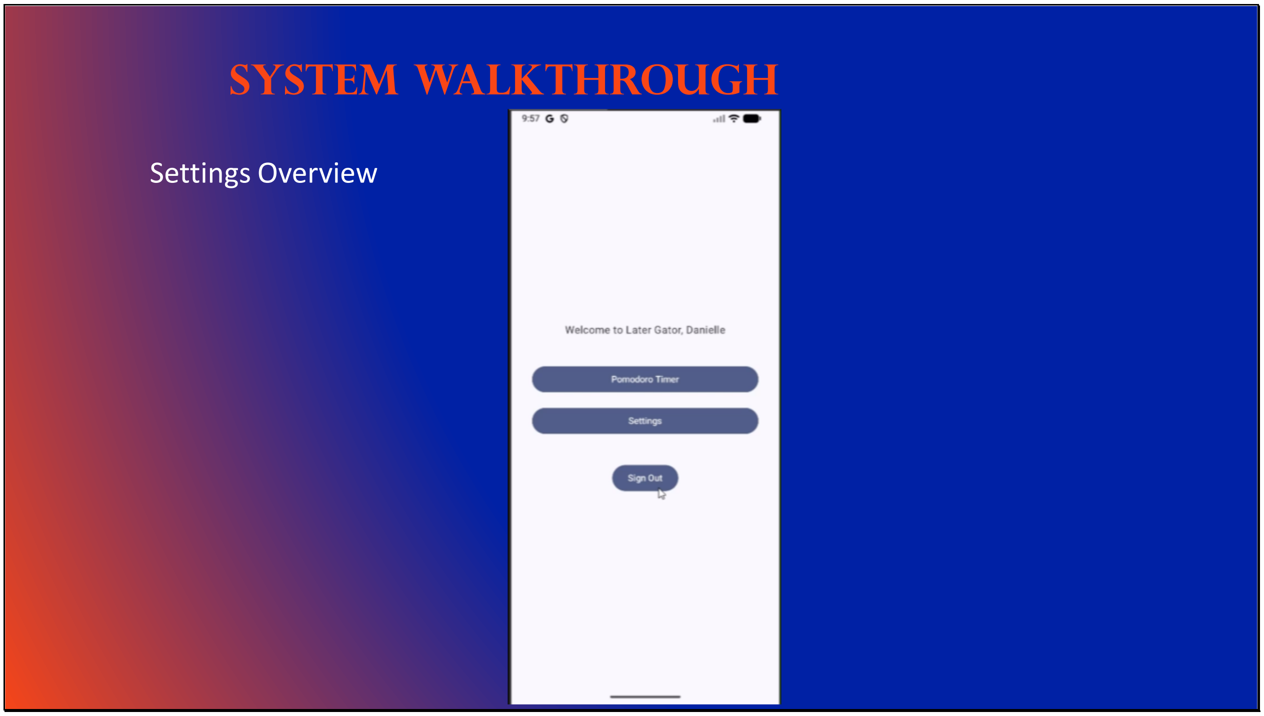
- Later Gator uses federated authentication, having users sign-in with Google
- User prompted to sign-in with a google account

Username creation

- Prompts user to enter a username so that a profile can be created
- The user can't enter a blank text box as their name
- Successfully creates the profile and brings user to the home screen

Now I'll walk through where the user can go from here
**NEXT SLIDE**

Slide 5




**DANIELLE:**

Log Out

- Signs the user out. Account already registered so user is brought straight to the home screen
  Update Username
- User can update username (just can’t be blank)

Permissions

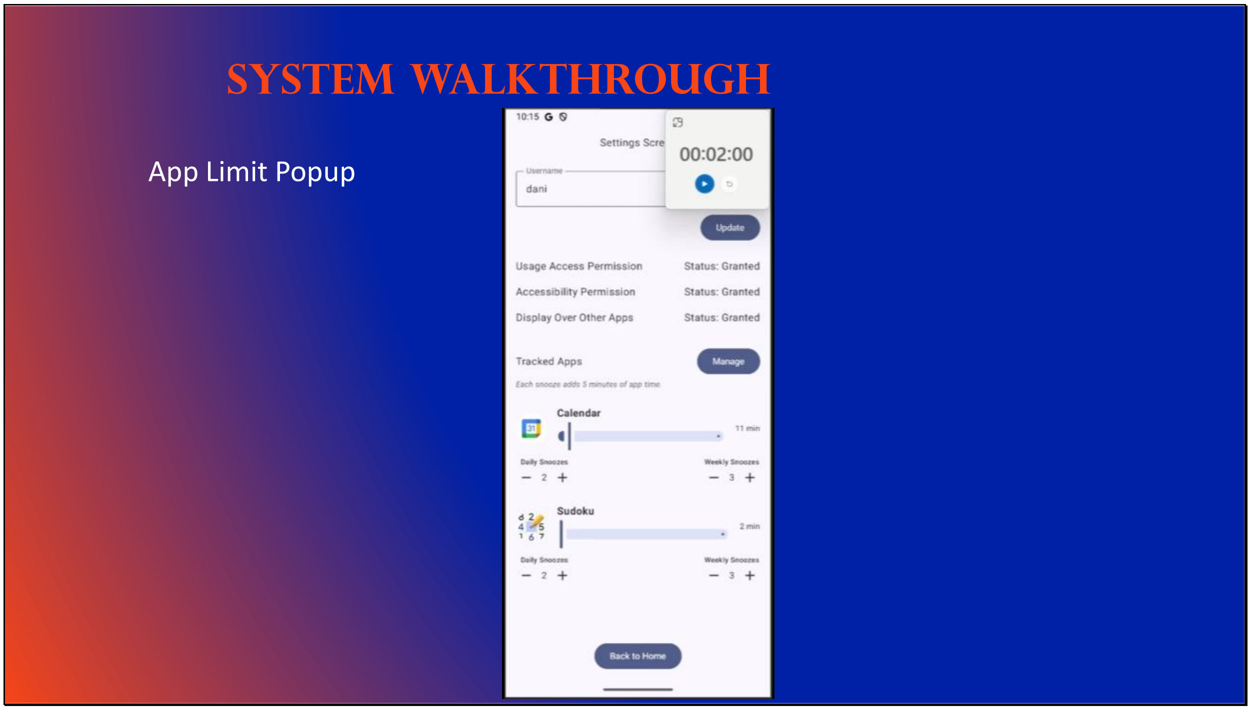
- Usage Access: Allows the app to see which other apps user is using and for how long. Needed for tracking
- Accessibility Service:  Allows the app to interact with the screen content and detect when a specific app is in the foreground. Needed to “block” an app
- Display Over Other App: Allows the app to draw a window on top of whatever else is on your screen. Needed to show the interruption screen when a limit is hit

App Selection

- Users select the ones they wish to track, and they will appear on the main settings screen
- When apps are first selected, they do not have a time limit set. User moves slider to set a limit
- Once limit is set, user can set snooze allowances per app. Weekly >= daily

Next, I’ll walk you through what it looks like when a limit is reached
**NEXT SLIDE**

Slide 6



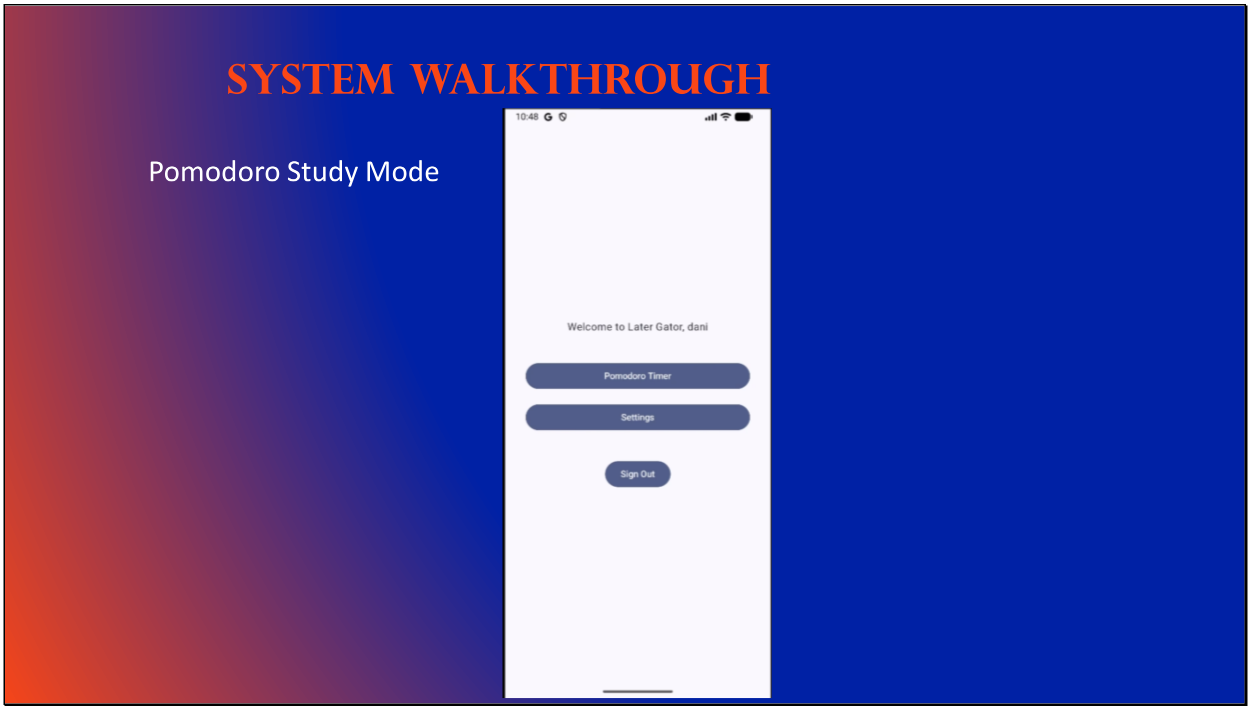

**DANIELLE:**

App Limit Popup Experience

- Sudoku is being tracked with a 2 min limit and 2 snoozes granted.
- Can Snooze or Exit App
- Snooze grants the user 5 more minutes on the app
- No more snoozes available, user must exit app

Now, let’s look at the Pomodoro study feature **NEXT SLIDE**

Slide 7



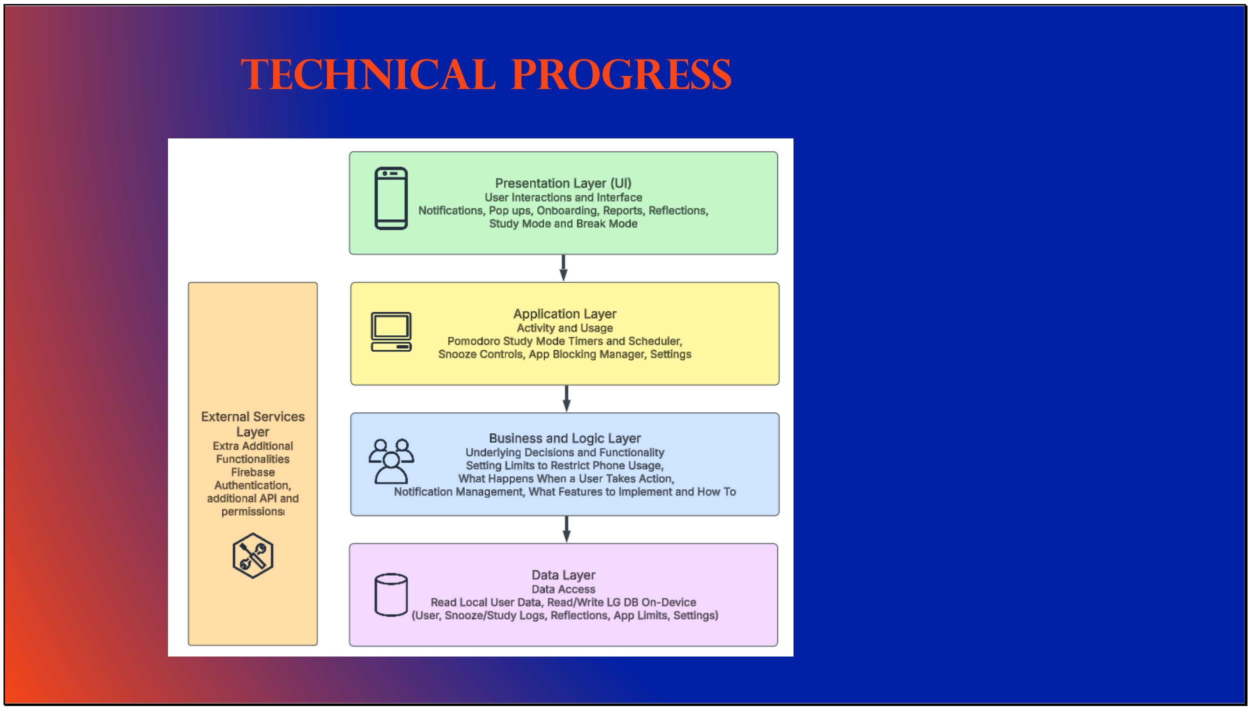

**DANIELLE:**

Pomodoro Study Mode

- 25 min study, 5 min break
- Could restart the study if you wanted
- When sessions are complete, return to home screen
- Can also click the return home at any time to return

Now to Carrie for our Technical Progress Pass **to Carrie**
**NEXT SLIDE**

Slide 8



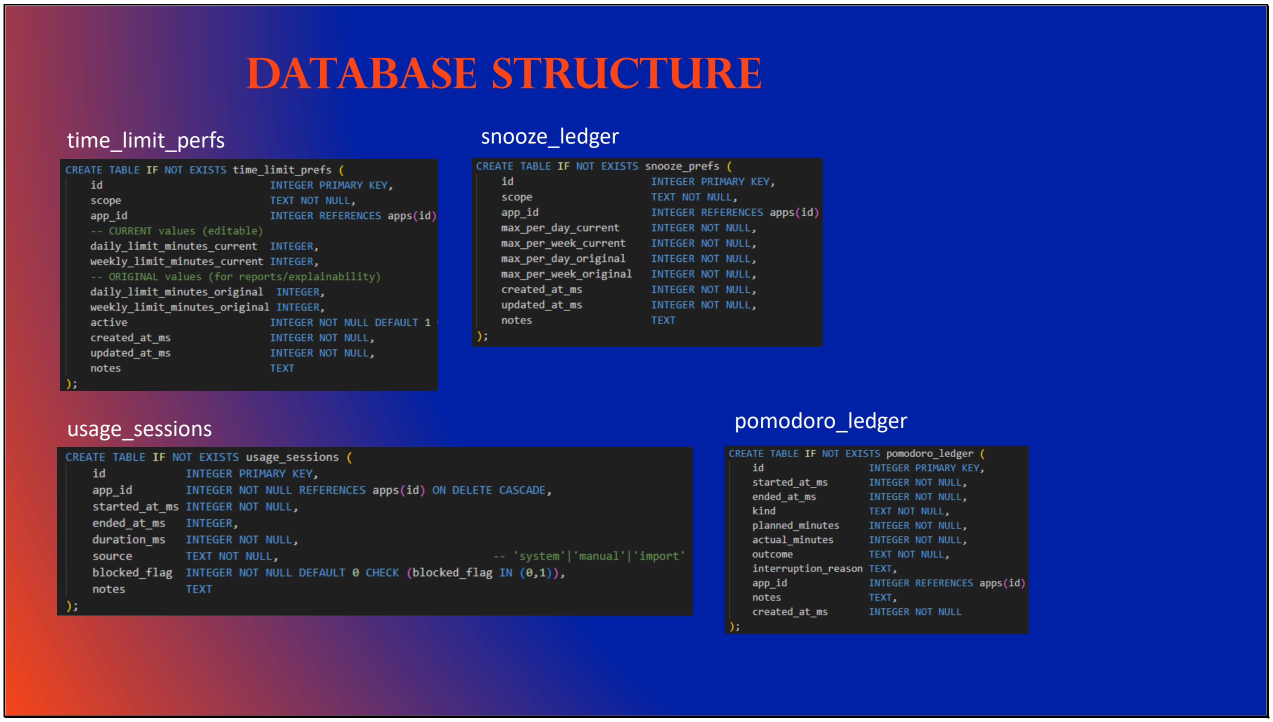

**CARRIE:**

Here's a quick reminder of our system model. We updated this version to reflect the final architecture we used during development. The layers you see here match the structure in our Documentation Report, and they helped guide how we organized our code.

The Data Layer is shown at the base, and that really is the foundation of the entire app. Everything else depends on being able to reliably store and retrieve user behavior

**NEXT SLIDE**

Slide 9




**CARRIE:**

We included over thirty tables in this prototype release. We know that sounds like a lot, but doing that work upfront really sets us up for faster development in future sprints because the structure already exists for more advanced features.

Here are just a few of the key tables we relied on:

- The time_limit_prefs table stores both the daily and weekly limits a user sets for each tracked app. This also includes the original values for reporting and explainability later on.

- The usage_sessions table tracks active foreground app sessions. This is what tells us how long a user has been in a particular app.

- The snooze_ledger table tracks every time a user chooses to snooze a limit instead of exiting an app.

- And the pomodoro_ledger stores work and break intervals for Study Mode. This will help us build out reports on productivity and focus sessions.

These tables make up the core of how Later Gator understands user behavior and how it can generate summaries, enforce limits, and adapt to the user's actions.

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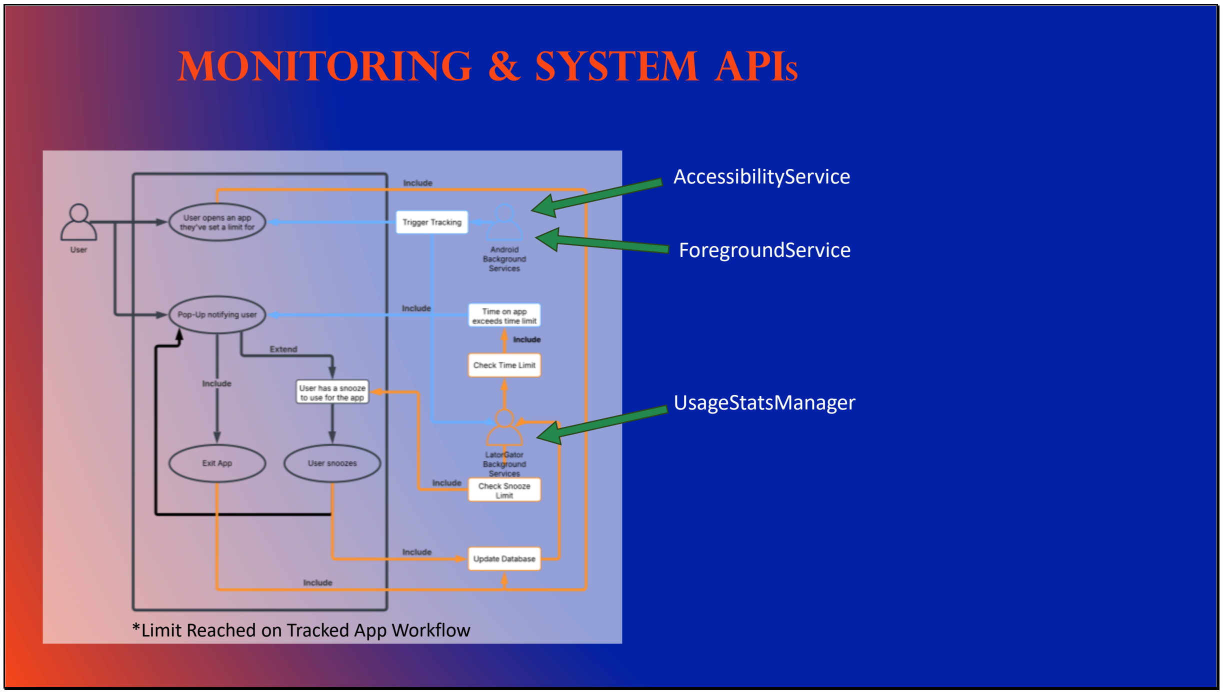
Slide 10




**CARRIE:**

Now I'll explain the system-level components that make our monitoring possible.
Danielle already showed the permissions required for these, so here's how they function under the hood.

- AccessibilityService allows us to capture which app is currently in the foreground. That information is essential because the app can't enforce limits unless it knows what the user is doing at any given moment.

- UsageStatsManager helps us retrieve usage history, which we then store in the database for reporting. This is particularly important when a user switches quickly between apps.

- And the Foreground Service keeps our monitoring running even when the app itself isn't open. That's how we continue tracking activity in real time and check whether limits have been reached.

These system APIs allow us to detect activity, enforce limits, log sessions, and update the database as the user moves throughout their device.

**NEXT SLIDE**

Slide 11


**CARRIE:**

For this release, we made very intentional tradeoffs.
We prioritized delivering a working prototype with core functionality, and that meant sacrificing some feature goals in exchange for stability.

One of the biggest areas of technical debt is in our DatabaseHelper file, which you can see scrolling on the screen. It currently holds a lot of responsibility, and the next major engineering task would be to break that into smaller components. This will improve readability, maintainability, and testability.

We also need to enforce consistent SQL patterns and expand our commenting.

Those improvements will make it much easier to add new features without introducing bugs.
These are the parts of the system we intentionally postponed so we could deliver stable functionality first, and they'll be the first things we address should we decide to continue work on this project after this semester.

Now Madison will report on our progression through our final sprint and future goals for the project.

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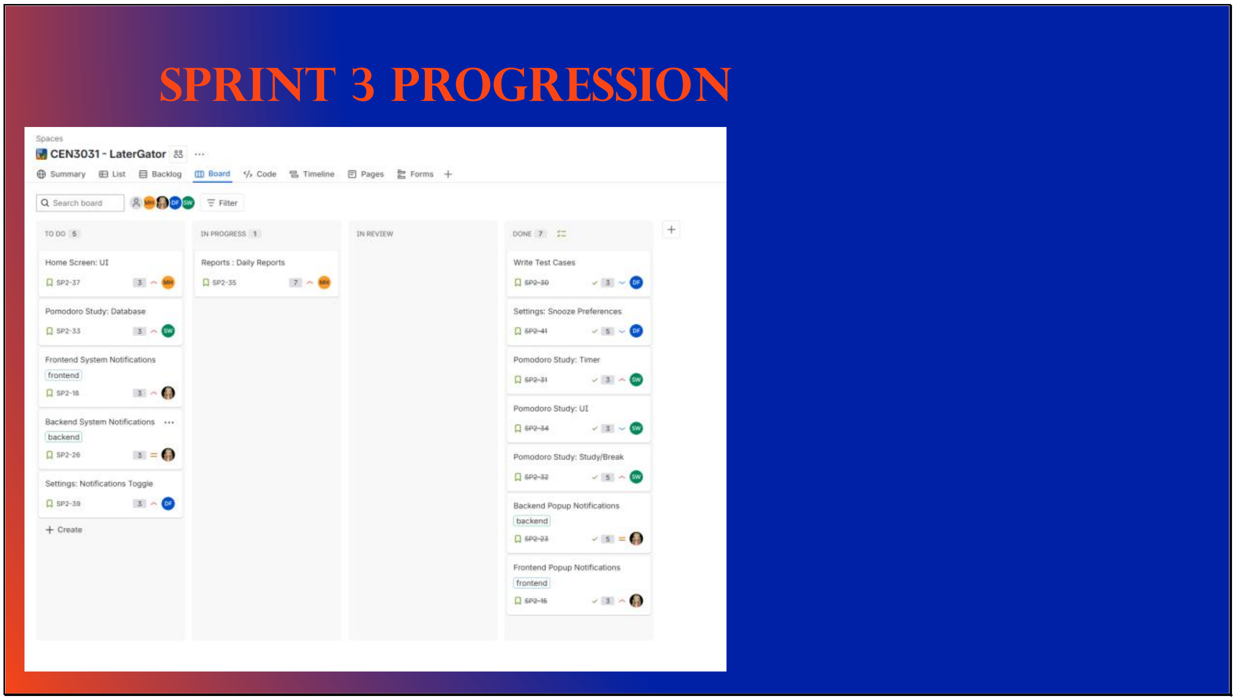
Slide 12




**MADISON:**

As we moved into this sprint, our team finally had a stable development environment in Android Studio, which made a huge difference in our workflow. With everything properly configured, we were able to make solid progress toward our core features. This sprint, we successfully implemented App Time Limit Monitoring, popup notifications, and our Pomodoro Study Timer. These features required a lot of debugging, trial and error, and collaboration, but they taught us how to work through a real development pipeline together instead of just in theory.

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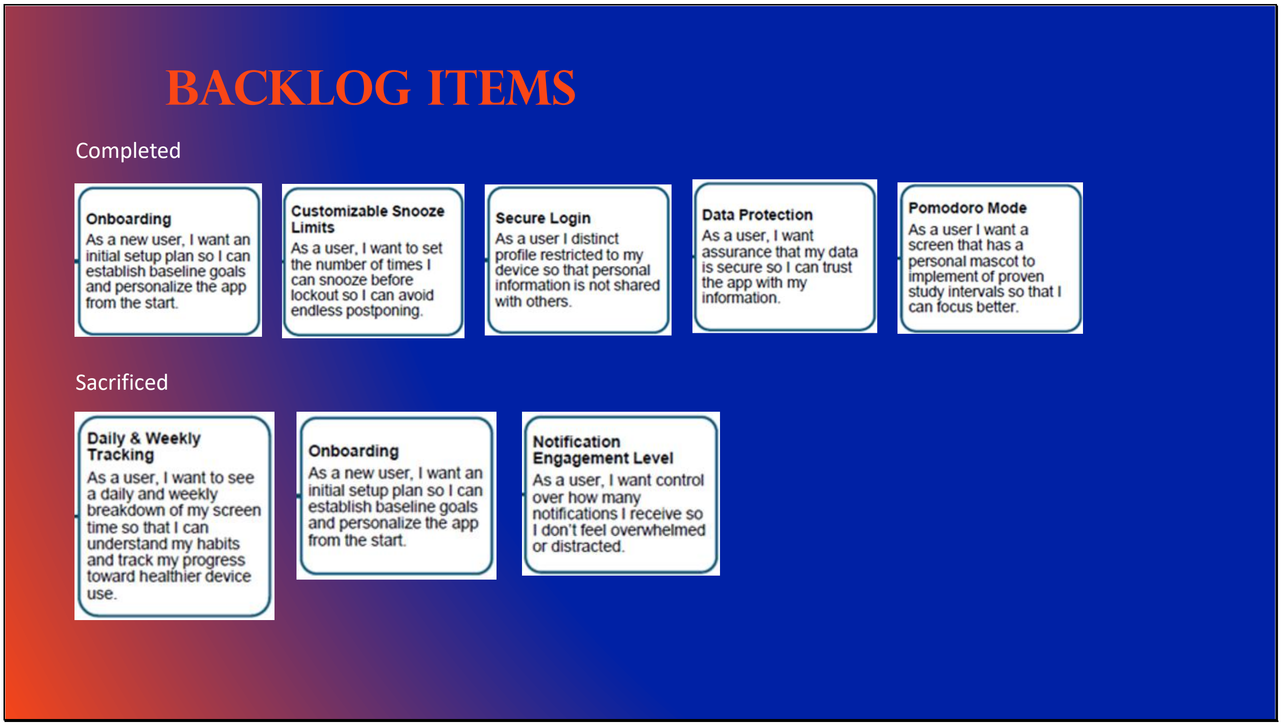
Slide 13




**MADISON:**

We weren't able to complete every task we initially planned, but instead of seeing that as a failure, we approached it the way AGILE intends, by reassessing, adapting, and reprioritizing. Our completed backlog items ended up reflecting a more realistic understanding of our skill level and our time. We learned to manage scope, to avoid overcommitting, and to make room for unexpected challenges.

Even though we left some items unfinished, we still moved the project forward in a meaningful way, and that's a win for us as junior developers learning this process.

**NEXT SLIDE**

Slide 14


**MADISON:**

One positive outcome of not completing everything is that we now have a very strong set of stretch goals. All the features that we originally imagined during our Blue Sky phase still align with our overall vision for the app. Because we've now built the solid foundations of the MVP, we have a clearer sense of how these stretch features can fit in. These include elements that make the app more engaging, more personalized, and more useful to real users. They represent the direction we want our project to grow toward, and they give us a roadmap for future development beyond the class. Our work proves we can actually *build* the things we set out to build. Even though we ran into all the classic junior developer struggles, environment issues, underestimated timelines, debugging nightmares, we overcame them as a team. Every setback made us more confident in how to collaborate, how to adapt our sprint planning, and how to break work down into achievable pieces.

Our stretch goals aren't just wish-list items. They represent the next logical steps forward for a team that now has real experience with navigating AGILE, Android Studio, and app development. We're finishing the semester not just with an MVP, but with a strong foundation and a clear vision for what comes next.

Pass off to Sam.

**NEXT SLIDE**

Slide 15


**SAM:**

We're all invested in Later Gator, and we hope to continue building it beyond this course. This project challenged us, stretched us, and genuinely helped us grow as developers.

We are proud of what we built this semester, thank you for the opportunity to learn, collaborate, and create something meaningful.